Census Income Analysis & Prediction

Exploratory Data Analysis (EDA) & Machine Learning Classification

Project Weightage: 50% | Dataset: UCI Adult (48,842 rows × 14 features) | Team: Dhruven • Prit • Punit

1. Team Work Division & Deliverables

Dhruven

STAGE 1 & STAGE 2A: CLEANING & UNIVARIATE

- **Notebook:** `01_dhruven_cleaning_univariate.ipynb`
- Data screening (types, shape, missing values)
- Clean `" ?"` missing values and duplicates
- Numerical & categorical boundary checks
- Imputation logic (categorical & numeric)
- Univariate distribution plots (Histograms + KDE)
- Income class distribution visualization
- Export cleaned CSV to `data/processed/`

Prit

STAGE 2B: TRANSFORMATIONS & BIVARIATE EDA

- **Notebook:** `02_prit_transformation_bivariate.ipynb`
- Outlier detection & Boxplot visualizations
- Skewness handling (Log transform of capital gain/loss)
- Categorical Encoding (One-Hot & Label Encoding)
- Target binary mapping (`<=50K` → 0, `>50K` → 1)
- Bivariate analysis (Education vs Income, Age vs Income)
- Multivariate correlation matrix & heatmap
- Export encoded feature matrix for modeling

Punit (Lead)

STAGE 3 & STAGE 4: ML MODELS & REPORTING

- **Notebook:** `03_punit_model_building.ipynb`
- Stratified 80/20 train-test split & feature scaling
- Train 4 models: Logistic Regression, Decision Tree, Random Forest, KNN
- Evaluation metrics: Accuracy, Precision, Recall, F1, ROC-AUC
- Confusion Matrices & ROC Curves comparison
- Feature Importance plot (Key income drivers)
- Master Model Comparison Summary Table
- Stage 4 executive summary & project presentation

2. Repository Structure Guidelines

To prevent Jupyter Notebook merge conflicts, each team member works strictly in their own assigned notebook inside `notebooks/`:

```
census-income-eda-ml/
├── data/
│   ├── raw/                # Original raw dataset (e.g. adult.csv)
│   └── processed/          # Cleaned & encoded dataset exports
├── notebooks/
│   ├── 01_dhruven_cleaning_univariate.ipynb   # Dhruven: Cleaning & Univariate EDA
│   ├── 02_prit_transformation_bivariate.ipynb  # Prit: Outliers, Encoding & Bivariate EDA
│   ├── 03_punit_model_building.ipynb           # Punit: ML Model Training & Evaluation
│   └── final_master_project.ipynb              # Merged master submission notebook
├── models/                 # Saved model artifacts (.joblib / .pkl)
├── reports/
│   ├── figures/            # High-resolution exported charts (ROC curves, heatmaps)
│   └── Census_Income_Project_Work_Division.pdf
├── .gitignore              # Ignores checkpoints, temp files, virtual environments
├── README.md               # Master project documentation with badges & credentials
└── requirements.txt        # Project dependencies (pandas, scikit-learn, seaborn, etc.)
```

3. Git Collaboration Workflow (How to Push Without Conflicts)

Step 1: Clone the Repository
`git clone https://github.com/punit169/census-income-eda-ml.git` → `cd census-income-eda-ml`

Step 2: Create and Switch to Your Personal Branch
Dhruven: `git checkout -b feature-dhruven-cleaning`
Prit: `git checkout -b feature-prit-eda`
Punit: `git checkout -b feature-punit-modeling`

Step 3: Work strictly in your assigned notebook
Do NOT modify another member's notebook file directly. Place generated charts in `reports/figures/`.

Step 4: Commit and Push to Your Branch
```
git add .
git commit -m "Completed [describe what you worked on]"
git push origin <your-branch-name>
```

Step 5: Pull Request & Review
Open a Pull Request on GitHub to merge into `main`. Punit will review and merge into the main branch.

4. Scoring & Evaluation Checklist (Targeting 50/50 Marks)

Stage	Key Criteria Required by Professor	Owner	Status
Stage 1	Problem definition, dataset source (UCI catalog link), attribute metadata	All Members	✓ Done
Stage 2A	Explicit `" ?"` handling, duplicate removal, boundary validation, imputation, univariate distribution plots	Dhruven	In Progress
Stage 2B	Outlier boxplots, log-transform for skewness, One-Hot Encoding, target binary mapping, bivariate plots & correlation heatmap	Prit	Ready to start
Stage 3	80/20 Stratified split, StandardScaler, 4 ML models, Confusion Matrices, ROC curves, Model Comparison Summary Table	Punit	Ready to start
Stage 4	Feature importance analysis, socioeconomic insights, final conclusions, clean report presentation	Punit + Team	Pending